

Tazkarti ASP.NET Core Web API

Architecture Design & Implementation Specification | Onion Architecture • .NET 10 • EF Core • SQL Server

Executive Summary: This document specifies the comprehensive backend API architecture for the Tazkarti Sports & Entertainment Ticketing Platform. The design uses **Onion Architecture** with strict separation of concerns across Domain, Application, Infrastructure, and API layers, leveraging ASP.NET Core (.NET 10), Entity Framework Core with Microsoft SQL Server, and ASP.NET Identity with custom **14-digit Egyptian National ID** primary keys.

CRITICAL REQUIREMENT — 14-Digit Egyptian National ID as PK

The AppUser entity inherits from IdentityUser<string> where the primary key (Id) is constrained strictly to 14 numeric digits matching the Egyptian Civil Registry standard. A SQL Server check constraint enforces LEN(Id) = 14 AND Id NOT LIKE '%[^0-9]%'

DATABASE CONNECTION & ENVIRONMENT SPECIFICATION

Target Server: YASSIN-PC\MSSQLSERVER01 | **Database:** TazkartiDb
Connection String: server=YASSIN-PC\MSSQLSERVER01;database=TazkartiDb;trusted_connection=true;TrustServerCertificate=True
Target Framework: .NET 10.0 (C# 13) | **Auth:** Identity Core (JWT Bearer to be integrated later)

1. Solution Architecture & Onion Layers

```
d:\TazkartiApi\  
■■■ Tazkarti.slnx (Solution File)  
■■■ src/  
    ■■■ Tazkarti.Domain/ (Core domain - zero dependencies)  
        ■ ■■■ Entities/ (AppUser, Club, Match, Event, BookingOrder, etc.)  
        ■ ■■■ Enums/ (LoyaltyTier, BookingStatus, PaymentMethod, etc.)  
        ■ ■■■ Interfaces/ (IGenericRepository<T>, IUnitOfWork)  
    ■■■ Tazkarti.Application/ (Use cases, contracts & business interfaces)  
        ■ ■■■ Interfaces/ & DTOs/  
    ■■■ Tazkarti.Infrastructure/ (Data persistence, EF Core, Identity, Repositories)  
        ■ ■■■ Data/ (TazkartiDbContext & 13 Fluent API Configurations)  
        ■ ■■■ Repositories/ (GenericRepository<T>, UnitOfWork)  
        ■ ■■■ DependencyInjection.cs (Service registration extension methods)  
    ■■■ Tazkarti.Api/ (ASP.NET Core Web API host, Controllers, Middleware)  
        ■■■ Controllers/  
        ■■■ Program.cs & appsettings.json
```

2. Domain Entities & Database Schema (13 Entities)

Extracted directly from the Tazkarti front-end implementation:

Entity	Key Properties & Data Types	Relationships & Navigation
AppUser (IdentityUser<string>)	Id (PK, 14-digit National ID) , FanId (TZK-YYYY-XXXX), FullName, DateOfBirth, Gender, Nationality, Governorate, AvatarUrl, QrCodeUrl, LoyaltyTier, AttendancePoints (500+), IsVerified, CreatedAt	1-to-N: Bookings, Payments, AttendanceRecords
Club	Id (PK, int) , Name, ShortName (3-4 char code), LogoUrl, PrimaryColor (Hex #RRGGBB)	1-to-N: HomeMatches, AwayMatches
StadiumVenue	Id (PK, int) , Name, Capacity, Location, City, MetroAccess, ImageUrl, Description, TurnstilesOpeningHours, ProhibitedItems (JSON)	1-to-N: Gates, Matches, Events
VenueGate	Id (PK, int) , VenueId (FK), GateName, AllocatedFor	N-to-1: StadiumVenue
FootballMatch	Id (PK, int) , Title, Competition, Round, HomeTeamId (FK), AwayTeamId (FK), VenueId (FK), City, MatchDate, KickoffTime, GateOpenTime, AvailabilityStatus, AvailabilityPercent, MinPrice, BannerImage, IsActive	N-to-1: HomeTeam, AwayTeam, Venue 1-to-N: Categories, Bookings
MatchTicketCategory	Id (PK, int) , MatchId (FK), Name (CAT1/CAT2/CAT3/VIP), Price (decimal), Color, Available, GateAllocation	N-to-1: FootballMatch 1-to-N: Bookings
EntertainmentEvent	Id (PK, int) , Title, Category (Enum), Tag, Artist, EventDate, EventTime, VenueId (FK), City, MinPrice, BannerImage, Description, IsActive	N-to-1: StadiumVenue 1-to-N: Tiers, Bookings

EventTicketTier	Id (PK, int), EventId (FK), Name (Regular/Fan Pit/VIP/Royal), Price (decimal), Perks (JSON Array)	N-to-1: EntertainmentEvent 1-to-N: Bookings
BookingOrder	Id (PK, int), BookingReference (TKT-YYYY-XXXX), UserId (FK), BookingType, MatchId (FK?), EventId (FK?), MatchCategoryId (FK?), EventTierId (FK?), Venue, City, Gate, Block, TotalAmount, Quantity, PaymentMethod, Status, CreatedAt	N-to-1: AppUser, Match, Event, Category, Tier 1-to-N: TicketPasses 1-to-1: Payment
TicketPass	Id (PK, int), BookingOrderId (FK), CurrentFanId, OriginalFanId, HolderName, Row, SeatNumber, Price, Gate, Status, IsTransferred	N-to-1: BookingOrder 1-to-N: Transfers
TicketTransfer	Id (PK, int), TicketPassId (FK), SenderFanId, RecipientFanId, TransferredAt, OtpVerified, Status	N-to-1: TicketPass
PaymentTransaction	Id (PK, int), BookingOrderId (FK), UserId (FK), PaymentMethod, Amount, Currency (EGP), CardNumberMasked, CardExpiry, InstaPayAddress, MeezaNumberMasked, FawryCode, MobileWalletNumber, Status, CreatedAt	N-to-1: BookingOrder, AppUser
AttendanceHistory	Id (PK, int), UserId (FK), MatchTitle, Venue, AttendanceDate, PointsEarned, Status	N-to-1: AppUser

3. Domain Enums (11 Enums)

Enum Name	Allowed Values / Members	Usage Context
Nationality	Egyptian = 0, Other = 1	User Registration / Identification
Gender	Male = 0, Female = 1	User Demographics
LoyaltyTier	Silver = 0, Gold = 1, Platinum = 2	Fan Loyalty & Priority
AvailabilityStatus	HighAvailability, FastSelling, Available, AlmostFull, SoldOut	Match / Event Availability
BookingType	Match = 0, Event = 1	Order Classification
BookingStatus	Confirmed, Pending, Cancelled, Transferred, Attended	Order Lifecycle State
PaymentMethod	Card, InstaPay, Meeza, Fawry, Wallet	Egyptian Gateway Channels
PaymentStatus	Success = 0, PendingFawry = 1, Failed = 2	Transaction State
TicketStatus	Confirmed, Transferred, Attended, Cancelled	Turnstile Pass State
TransferStatus	Success = 0, Pending = 1, Rejected = 2	Pass Transfer Flow State
EventCategory	MusicAndConcerts, ClassicalAndOrchestra, RockAndIndie, ComedyAndTheater	Event Discovery

4. Infrastructure Layer & Pre-configured Seed Data

Seed Data included for Clubs & Venues:

- **9 Football Clubs:** Al Ahly SC (#b90015), Zamalek SC (#####), Pyramids FC (#00618d), Modern Sport FC (#1DA95E), Al Ittihad Alex (#AF7928), Ismaily SC (#e21e26), Egypt National Team (#CE1126), Senegal (#009639), Mamelodi Sundowns (#FFD700).
- **9 Stadiums & Arenas:** Cairo International Stadium (75k), Egypt Stadium New Capital (93.9k), Borg El Arab (86k), 30 June Stadium (30k), Alexandria Stadium (20k), New Alamein Arena (15k), Grand Egyptian Museum (10k), Zayed Central Park Arena (8k), The Marquee CFC (5k).

5. Target NuGet Packages & Versions (.NET 10)

Project	Package Name	Target Version	Purpose
Domain	Microsoft.Extensions.Identity.Stores	10.0.11	IdentityUser base entity
Infrastructure	Microsoft.AspNetCore.Identity.EntityFrameworkCore	10.0.11	Identity DbContext integration
Infrastructure	Microsoft.EntityFrameworkCore.SqlServer	10.0.11	SQL Server database provider
Infrastructure	Microsoft.EntityFrameworkCore.Tools	10.0.11	EF Core Migrations & CLI
Api	Swashbuckle.AspNetCore	6.6.2	Swagger OpenAPI UI Documentation

6. Execution & Verification Workflow

```
# 1. Build Entire Solution
dotnet build d:\TazkartiApi\Tazkarti.slnx

# 2. Add Initial EF Core Migration
dotnet ef migrations add InitialCreate --project d:\TazkartiApi\src\Tazkarti.Infrastructure --startup-project d:\TazkartiApi\src\Tazkarti.Api

# 3. Apply Migration & Create Database on SQL Server
dotnet ef database update --project d:\TazkartiApi\src\Tazkarti.Infrastructure --startup-project d:\TazkartiApi\src\Tazkarti.Api

# 4. Run API and Verify Swagger UI at http://localhost:5000/swagger
dotnet run --project d:\TazkartiApi\src\Tazkarti.Api
```